

Name : Tanishq J Ghodadra

Sem : 7th

Div : B

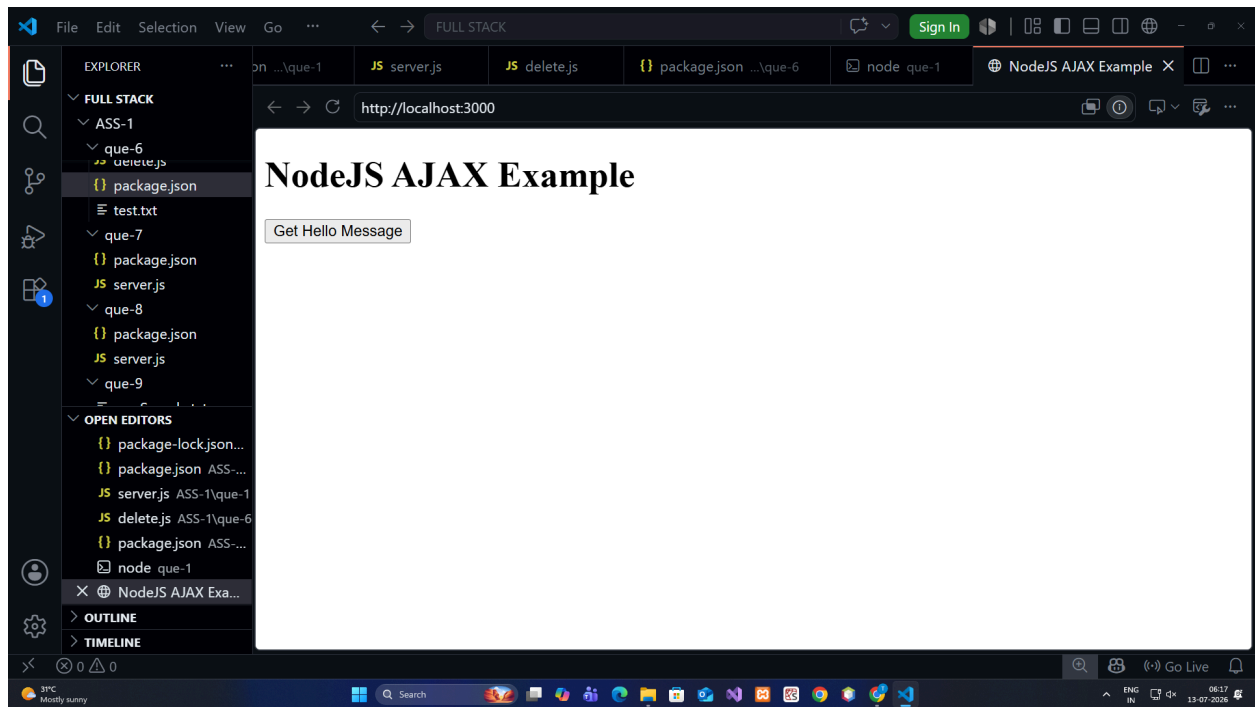
Roll No : 100

Subject : Full Stack

Que :1

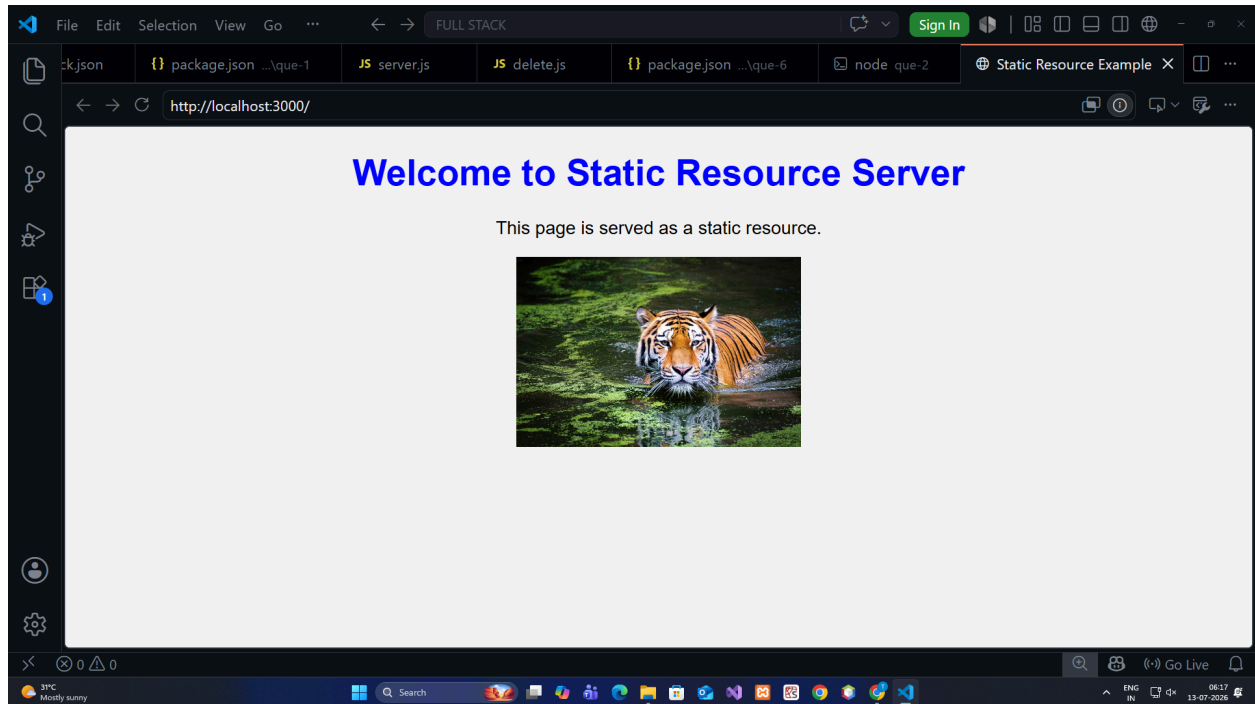
Develop nodejs application with following requirements:

- Develop a route `"/gethello"` with GET method. It displays "Hello NodeJS!!" as response.
- Make an HTML page and display.
- Call `"/gethello"` route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)



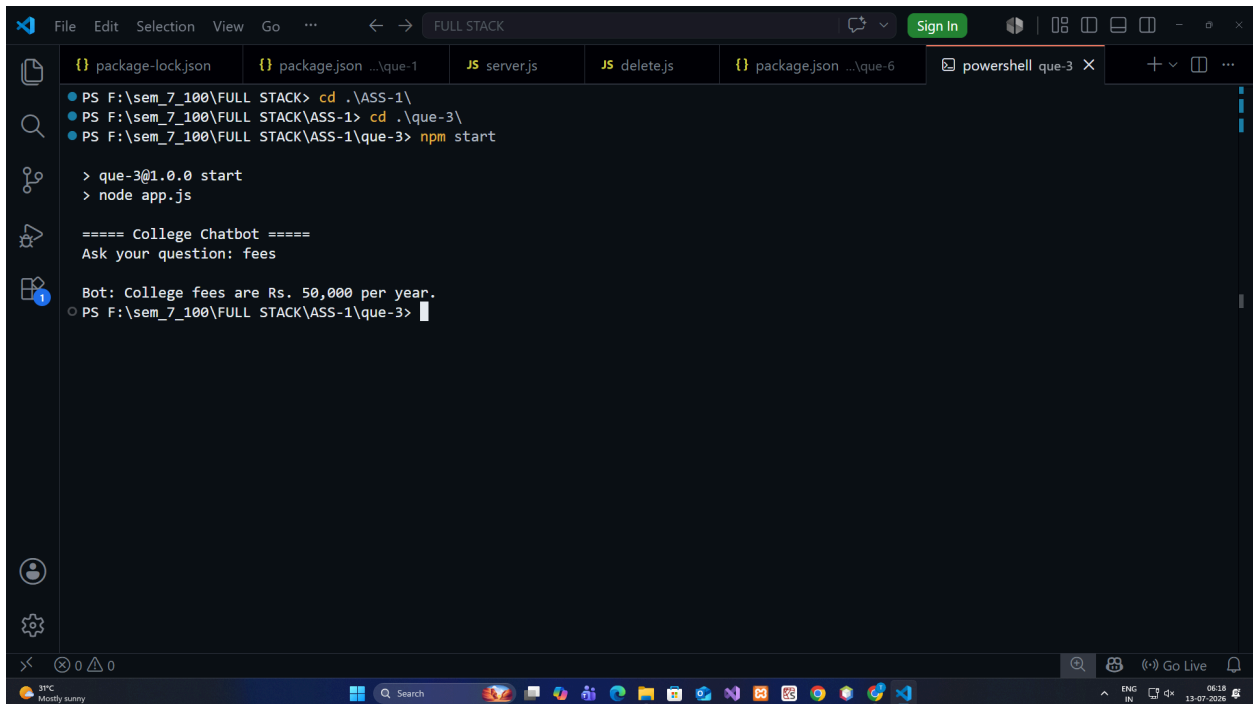
Que 2:

Develop a web server which serves static resources.



Que :3

Develop a module for domain specific chatbot and use it in a command line application.



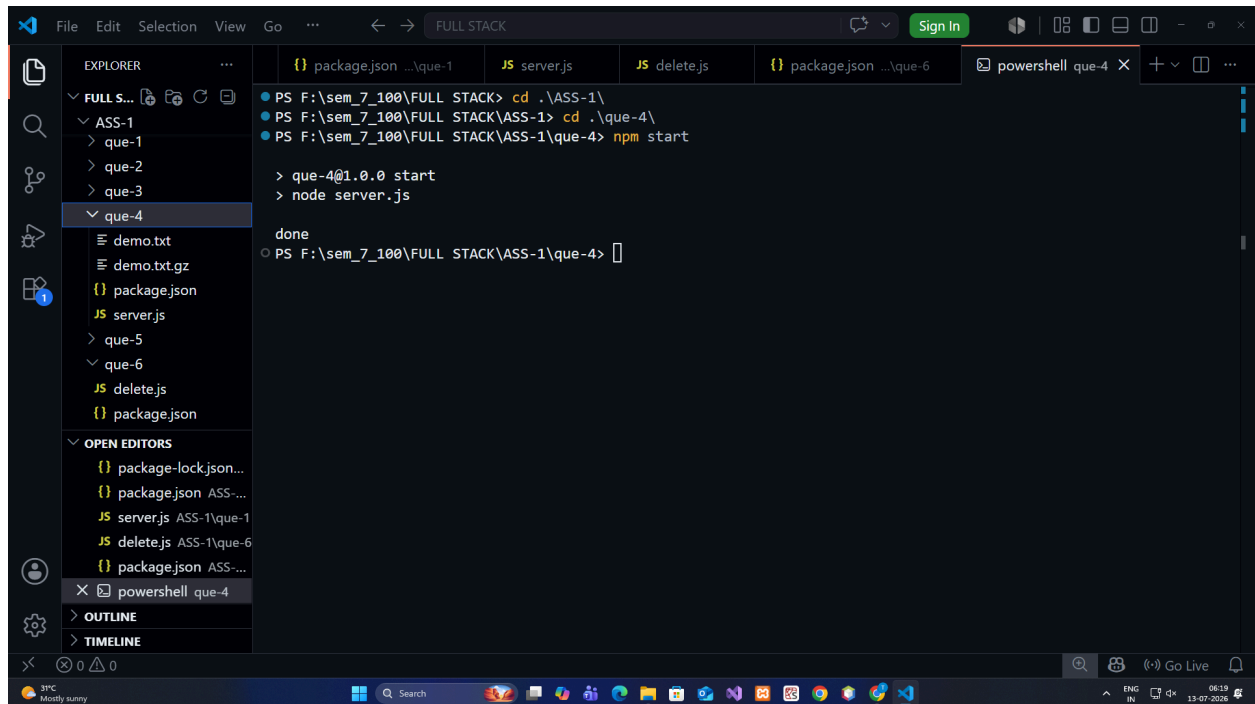
The screenshot shows a Visual Studio Code editor with a terminal window open. The terminal displays the following commands and output:

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-3\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-3> npm start  
  
> que-3@1.0.0 start  
> node app.js  
  
===== College Chatbot =====  
Ask your question: fees  
  
Bot: College fees are Rs. 50,000 per year.  
PS F:\sem_7_100\FULL STACK\ASS-1\que-3>
```

The terminal window is titled "powershell que-3" and shows the execution of the chatbot application. The chatbot responds to the question "fees" with the answer "College fees are Rs. 50,000 per year." The VS Code interface includes a sidebar with icons for Explorer, Search, Source Control, and Run and Debug. The bottom status bar shows the temperature as 31°C and the date as 13-07-2020.

Que :4

Write a program to create a compressed zip file for a folder.



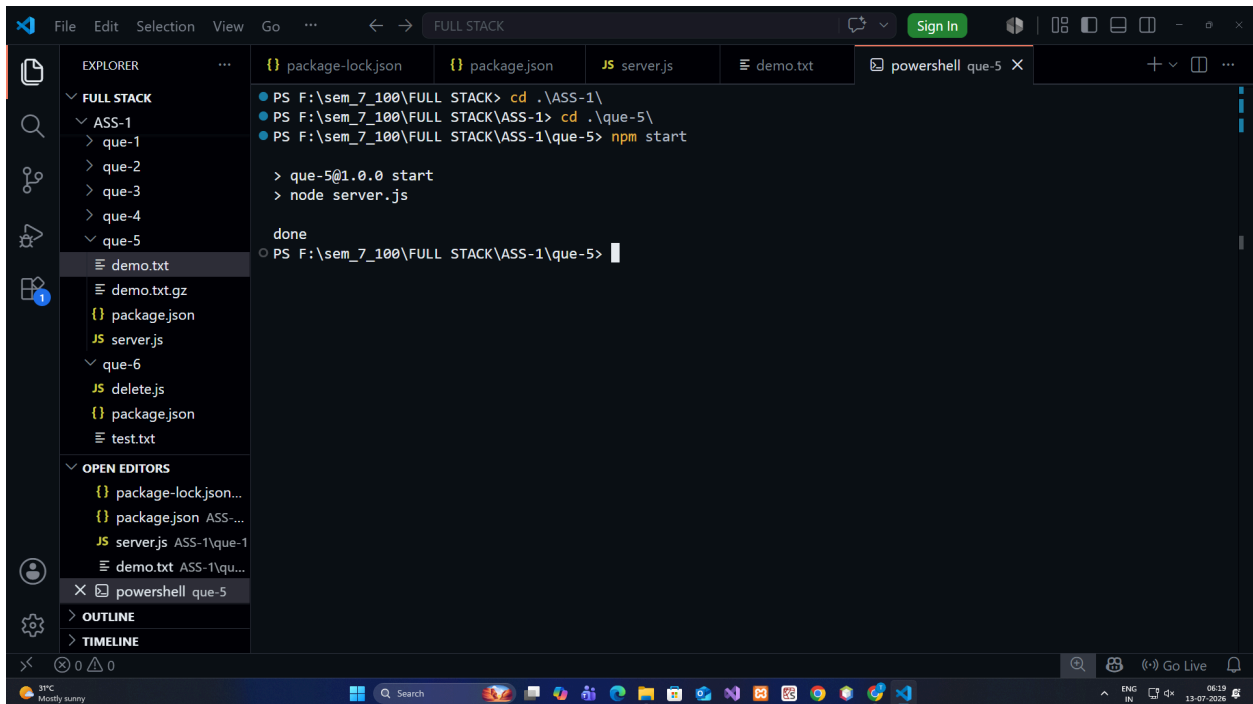
The screenshot shows the Visual Studio Code interface with a PowerShell terminal window open. The terminal displays the following commands and output:

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-4\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-4> npm start  
  
> que-4@1.0.0 start  
> node server.js  
  
done  
PS F:\sem_7_100\FULL STACK\ASS-1\que-4> 
```

The Explorer panel on the left shows the file structure of the project, including the 'que-4' folder and its contents: 'demo.txt', 'demo.txt.gz', 'package.json', 'server.js', 'que-5', 'que-6', 'delete.js', and 'package.json'. The Open Editors panel shows the files currently open in the editor.

Que :5

Write a program to extract a zip file.



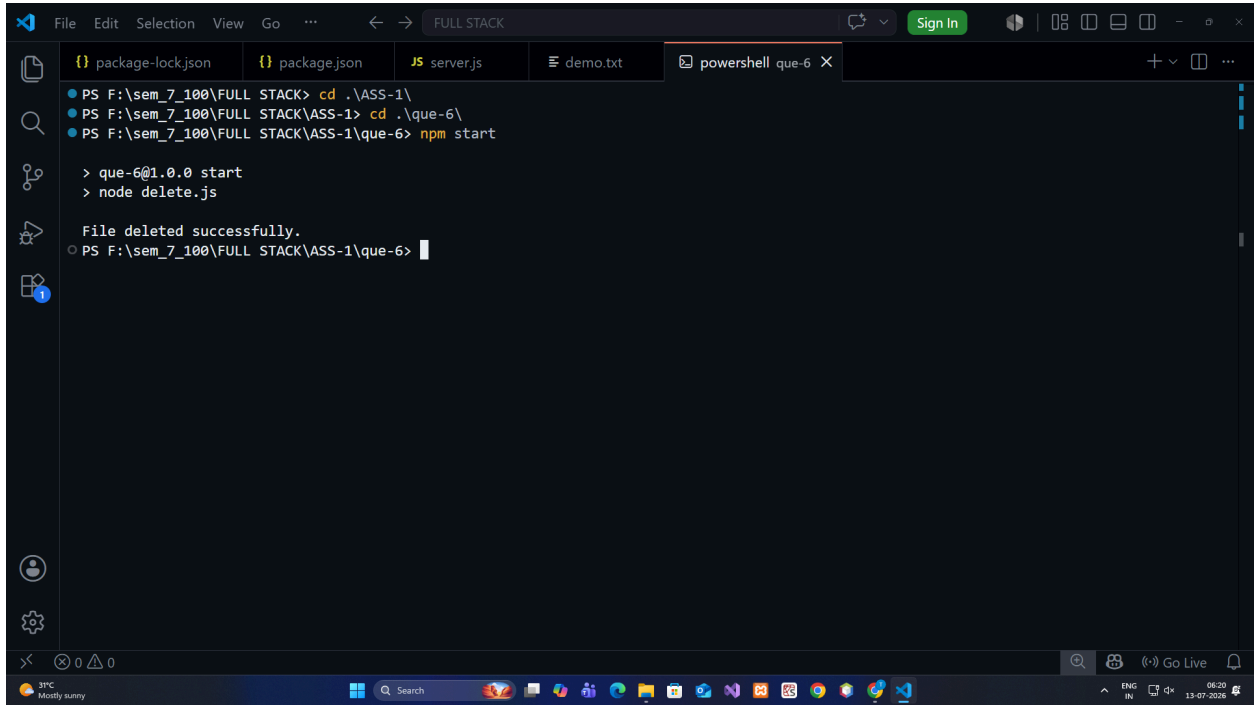
The screenshot shows the Visual Studio Code interface with a PowerShell terminal window open. The terminal displays the following commands and output:

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-5\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-5> npm start  
  
> que-5@1.0.0 start  
> node server.js  
  
done  
PS F:\sem_7_100\FULL STACK\ASS-1\que-5>
```

The Explorer sidebar on the left shows the file structure of the project, including folders like ASS-1, que-1, que-2, que-3, que-4, and que-5, and files like demo.txt, demo.txt.gz, package.json, server.js, delete.js, package.json, and test.txt. The Open Editors sidebar shows the currently open files, including package-lock.json, package.json, server.js, and demo.txt. The PowerShell terminal window is titled "powershell que-5".

Que :6

Write a program to promisify fs.unlink function and call it.



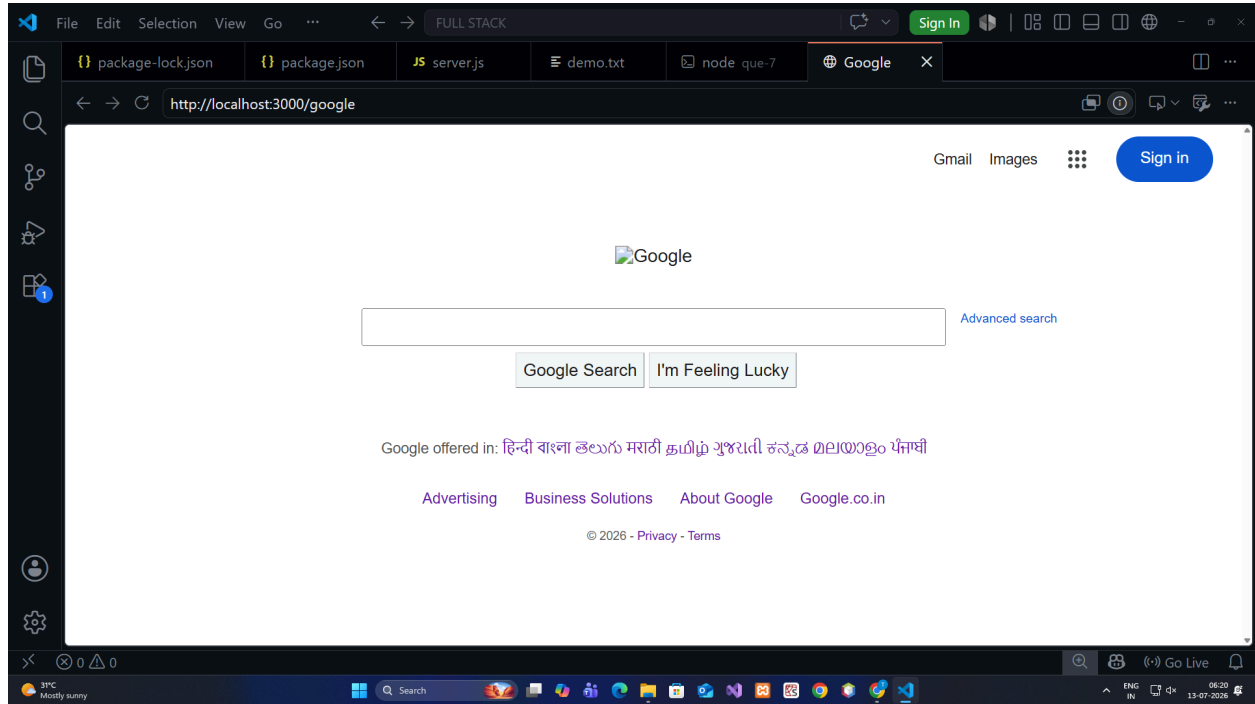
The screenshot shows a Visual Studio Code editor with a terminal window open. The terminal is running PowerShell commands in a directory structure. The commands and their outputs are as follows:

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-6\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-6> npm start  
  
> que-6@1.0.0 start  
> node delete.js  
  
File deleted successfully.  
PS F:\sem_7_100\FULL STACK\ASS-1\que-6>
```

The terminal window is titled "powershell que-6". The Visual Studio Code interface shows the "FULL STACK" workspace with files like package-lock.json, package.json, server.js, and demo.txt. The Windows taskbar at the bottom shows the date and time as 06:20 on 13-07-2026.

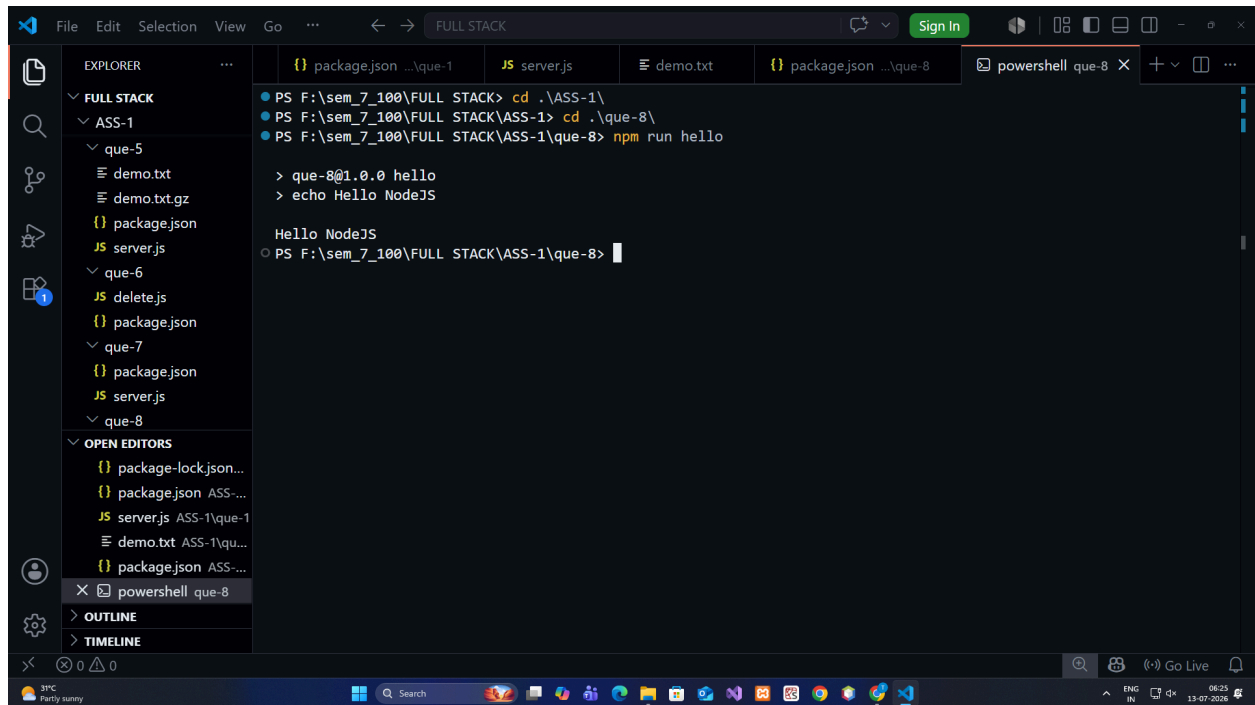
Que :7

Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.



Que :8

Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs application



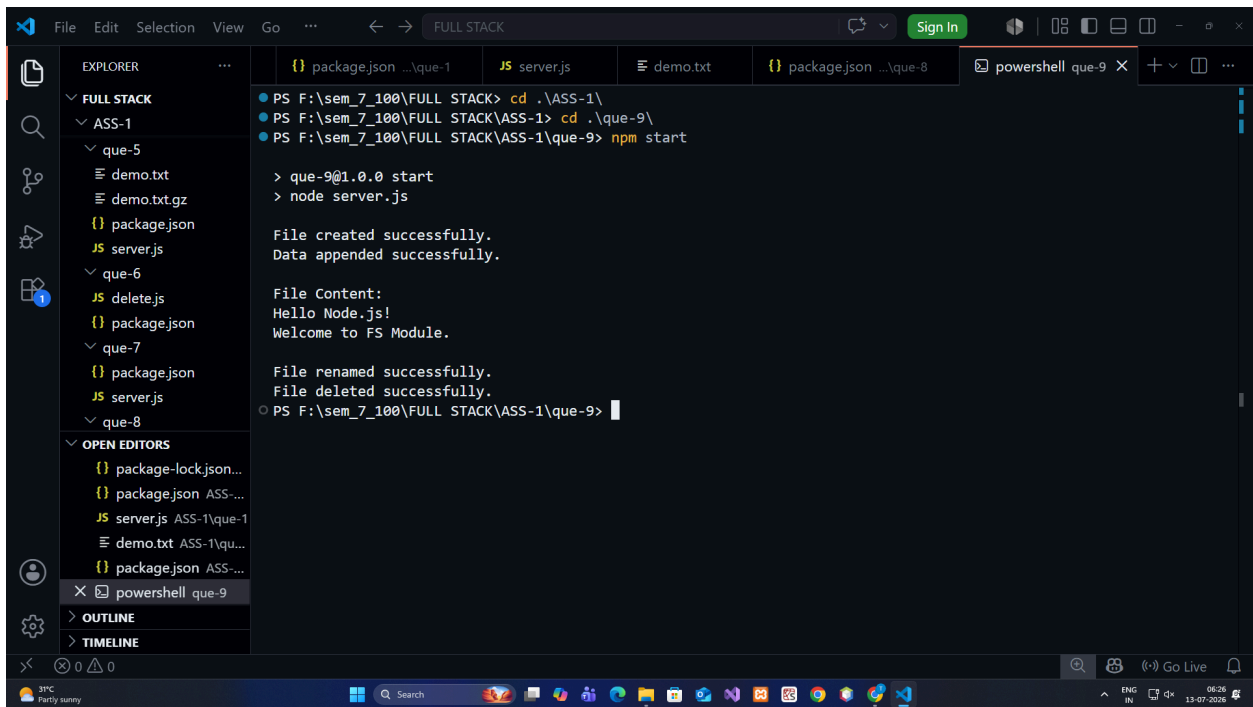
The screenshot shows the Visual Studio Code editor with the following details:

- Explorer Panel:** Displays a file tree with folders 'FULL STACK' and 'ASS-1'. Under 'ASS-1', there are subfolders 'que-5', 'que-6', 'que-7', and 'que-8'. Each folder contains files like 'demo.txt', 'demo.txt.gz', 'package.json', and 'server.js'. The 'que-8' folder is currently selected.
- Open Editors Panel:** Shows the 'package.json' file for the 'que-8' folder.
- Code Editor:** Displays the content of the 'package.json' file, which includes a 'scripts' section with the following entries:

```
"scripts": {
  "start": "node server.js",
  "test": "node test.js",
  "build": "node build.js",
  "run": "node run.js"
}
```
- Terminal Panel:** Shows the output of the 'npm run hello' command, which prints 'Hello NodeJS'.
- Bottom Panel:** Includes the 'OUTLINE' and 'TIMELINE' views.
- Taskbar:** Shows the Windows taskbar with various application icons and the system clock indicating 06:25 on 13-07-2026.

Que :9

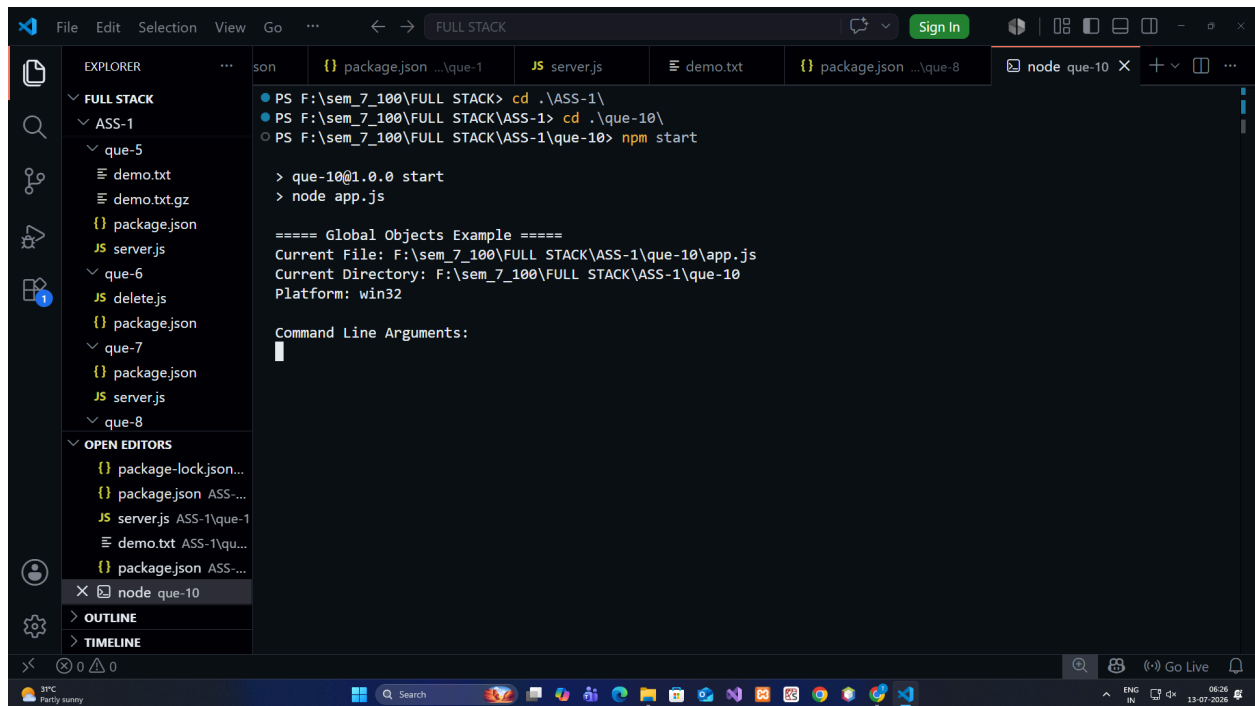
A program which calls useful functions in fs module.



```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-9\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-9> npm start  
  
> que-9@1.0.0 start  
> node server.js  
  
File created successfully.  
Data appended successfully.  
  
File Content:  
Hello Node.js!  
Welcome to FS Module.  
  
File renamed successfully.  
File deleted successfully.  
PS F:\sem_7_100\FULL STACK\ASS-1\que-9>
```

Que :10

A program which uses global objects in nodejs. Also read command line arguments and print it.



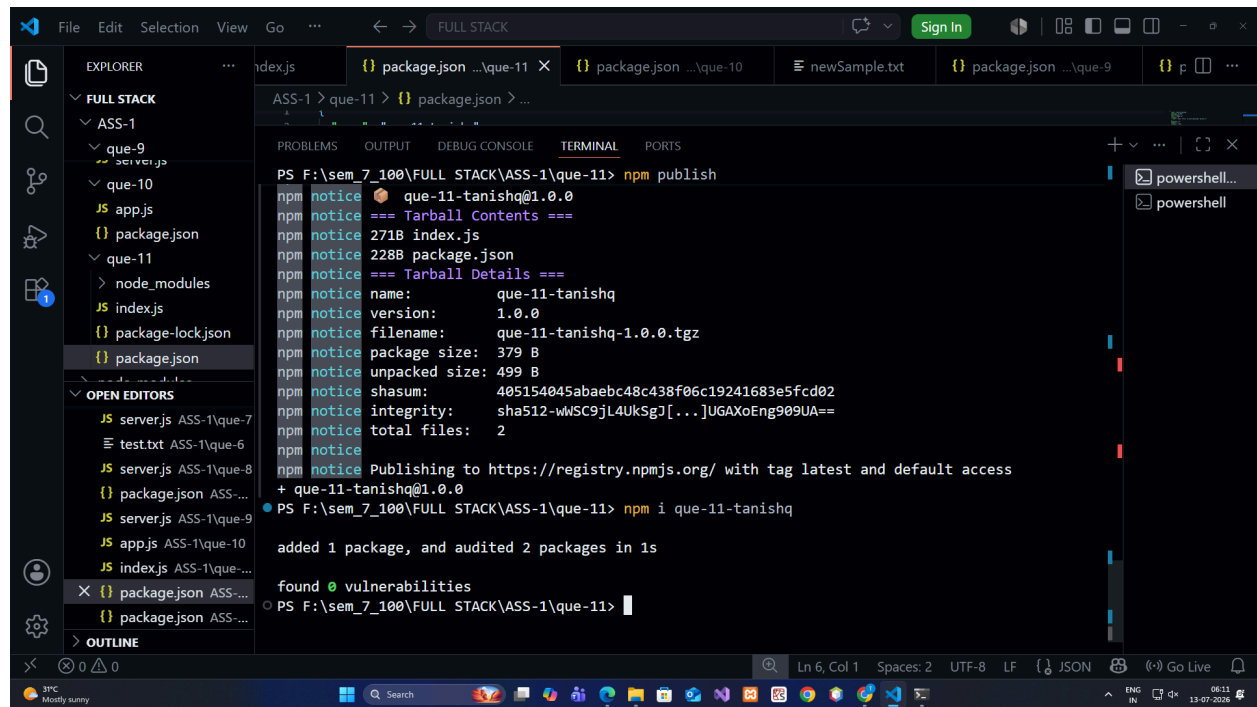
The screenshot shows the Visual Studio Code interface with a project named 'FULL STACK'. The Explorer sidebar on the left shows a folder structure under 'ASS-1' containing subfolders 'que-5' through 'que-8'. The 'que-10' folder is selected, showing files like 'demo.txt', 'demo.txt.gz', 'package.json', 'server.js', 'delete.js', and 'package.json'. The 'OPEN EDITORS' section shows the 'node que-10' editor is active. The main editor area displays a terminal window with the following content:

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-10\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-10> npm start  
  
> que-10@1.0.0 start  
> node app.js  
  
===== Global Objects Example =====  
Current File: F:\sem_7_100\FULL STACK\ASS-1\que-10\app.js  
Current Directory: F:\sem_7_100\FULL STACK\ASS-1\que-10  
Platform: win32  
  
Command Line Arguments:  
|
```

The bottom status bar shows the system clock as 06:26 on 13-07-2026, with the language set to English (IN) and the file encoding as UTF-8.

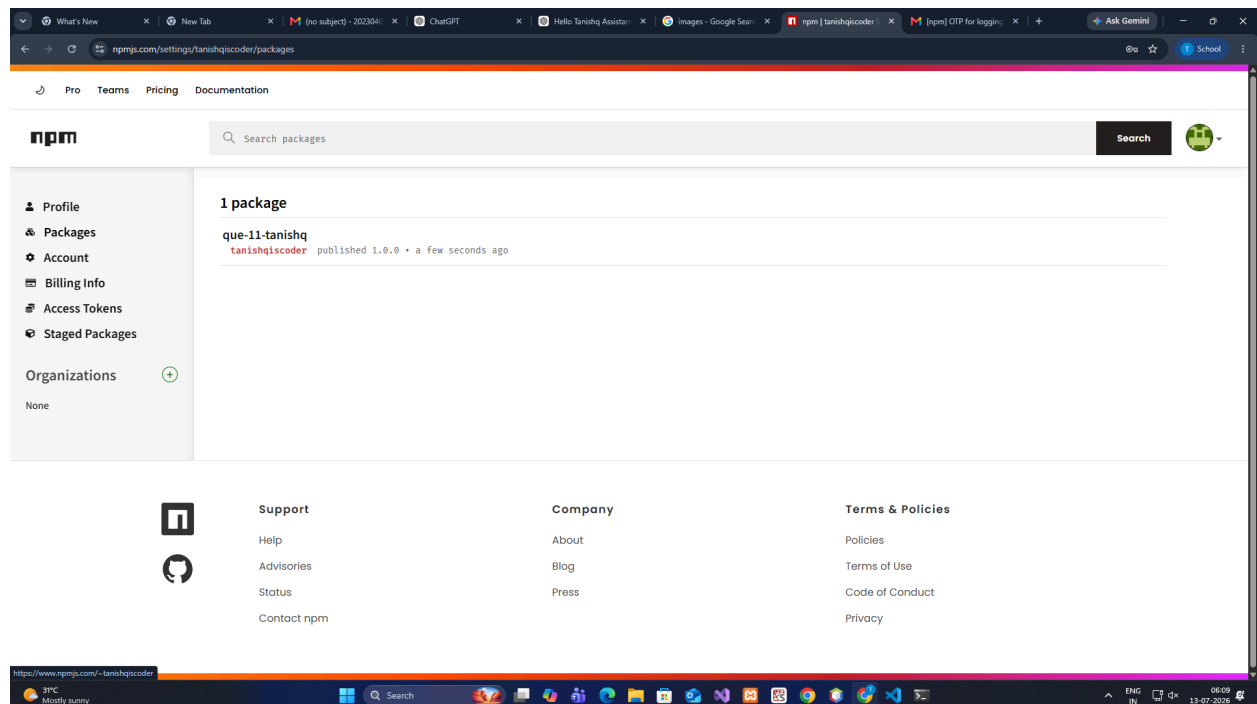
Que :11

Develop a useful package and publish it on npmjs.com. Write a program to use it.



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the command `npm publish` being executed in the directory `ASS-1\que-11`. The output shows the package details, including the name `que-11-tanishq`, version `1.0.0`, and the filename `que-11-tanishq-1.0.0.tgz`. The package is successfully published to the npm registry. The Explorer panel on the left shows the project structure, including `package.json`, `package-lock.json`, and `index.js`.

```
PS F:\sem_7_100\FULL STACK\ASS-1\que-11> npm publish
npm notice que-11-tanishq@1.0.0
npm notice === Tarball Contents ===
npm notice 271B index.js
npm notice 228B package.json
npm notice === Tarball Details ===
npm notice name: que-11-tanishq
npm notice version: 1.0.0
npm notice filename: que-11-tanishq-1.0.0.tgz
npm notice package size: 379 B
npm notice unpacked size: 499 B
npm notice shasum: 405154045abaebc48c438f06c19241683e5fcd02
npm notice integrity: sha512-wWSC9jL4UkSgJ[...]UGAXoEng909UA==
npm notice total files: 2
npm notice Publishing to https://registry.npmjs.org/ with tag latest and default access
+ que-11-tanishq@1.0.0
PS F:\sem_7_100\FULL STACK\ASS-1\que-11> npm i que-11-tanishq
added 1 package, and audited 2 packages in 1s
found 0 vulnerabilities
PS F:\sem_7_100\FULL STACK\ASS-1\que-11>
```



Que :12

Write a node program having at least 3 dependent tasks and call using chained promises.

The screenshot shows a Visual Studio Code editor with a project named 'FULL STACK'. The Explorer sidebar on the left shows a folder structure with 'ASS-1' containing subfolders 'que-5' through 'que-8'. The 'que-12' folder is selected, showing files 'demo.txt', 'demo.txt.gz', 'package.json', and 'server.js'. The 'server.js' file is open in the editor, displaying a Node.js script that uses chained promises to perform three tasks: creating a student, adding marks, and calculating a grade. The script also outputs the final student details as a JSON object.

```
PS F:\sem_7_100\FULL STACK> cd .\ASS-1\  
PS F:\sem_7_100\FULL STACK\ASS-1> cd .\que-12\  
PS F:\sem_7_100\FULL STACK\ASS-1\que-12> npm start  
  
> que-12@1.0.0 start  
> node server.js  
  
Task 1: Student Created  
Task 2: Marks Added  
Task 3: Grade Calculated  
  
Final Student Details:  
{ name: 'Tanishq', marks: 85, grade: 'A' }  
PS F:\sem_7_100\FULL STACK\ASS-1\que-12>
```